

LAUNCHLENS TECHNICAL SERIES

# MILESTONE 3: AI REASONING & LANGGRAPH AGENT WORKFLOWS

Detailed Technical Architecture, Autonomous Mitigation Engine,  
and Multi-Agent State Machine Documentation

## PROJECT SCOPE

Milestone 3 Implementation

## PRIMARY AI ENGINE

Google Gemini 3.6 Flash

## DOCUMENT VERSION

v1.0 (Production Final)

## TARGET AUDIENCE

Engineering & Architecture Team

## AGENT FRAMEWORK

LangGraph (@langchain/langgraph)

## STATUS

Verified & Operational

Confidential Technical Documentation — LaunchLens Full-Stack Platform

# 1. Executive Summary & Architecture Roadmap

Milestone 3 represents the core cognitive layer of the LaunchLens platform. While Milestone 1 provided structured data intake (projects, TAM/SAM/SOM market sizing, public stock quotes) and Milestone 2 established deterministic analytical scoring (5-category risk meters, SWOT matrix, 4-dimension feasibility models), **Milestone 3 transforms static metrics into dynamic, actionable intelligence.**

Milestone 3 is comprised of three interconnected, production-grade tasks built directly into the existing Express.js and React 19 architecture without disrupting existing features:

## Core Milestone 3 Modules:

- **Task 1 — AI-Powered Strategic Recommendations:** Multi-provider AI recommendation engine utilizing Google Gemini 3.6 Flash, OpenAI GPT-4o-mini fallback, and a zero-dependency deterministic rule engine fallback.
- **Task 2 — Mitigation & Improvement Suggestion Engine:** Problem identification engine analyzing context signals to synthesize root causes, tactical mitigations, product improvements, priority levels, actions, and expected outcomes.
- **Task 3 — LangGraph Multi-Agent Agent Workflow:** An 8-node state-machine graph built with @langchain/langgraph executing data analysis, risk detection, strategic synthesis, quality audit validation, and conditional routing.

## 2. Task 1: AI Strategic Recommendations

### 2.1 Engine Objectives & Design Constraints

Task 1 replaces generic static statements with data-grounded, contextual strategy cards. The AI engine consumes the complete intake context—including startup budget, business model, sector TAM growth, and category leader market share—to formulate customized recommendations.

#### Task 1 Provider Resolution & Fallback Sequence

Incoming Data Request → Check GEMINI\_API\_KEY → Call Gemini 3.6 Flash  
↳ If Gemini Fails / No Key → Check OPENAI\_API\_KEY → Call OpenAI GPT-4o-mini  
↳ If Both Fail / No Keys → Deterministic Rule Engine Fallback (Zero Error Downtime)

### 2.2 Data Grounding & Prompt Synthesis

To prevent AI hallucination and enforce strict relevance, prompt construction enforces explicit data binding. The model receives raw numeric inputs, such as SOM Growth (-1.2%) and Top Competitor (Flipkart, 31% share), requiring every generated item to provide a rationale explaining why the strategy directly addresses the project's empirical metrics.



### 2.3 Backend Service Architecture (aiService.js)

The backend service `backend/src/services/aiService.js` implements a resilient, multi-model execution function titled `callGeminiApi()`. To ensure model availability across regional Google AI Studio accounts, it cycles through candidate endpoints (`gemini-3.6-flash`, `gemini-3.5-flash`, `gemini-2.5-flash`, `gemini-1.5-flash`) using both the official `@google/genai` SDK and native REST API calls.

```
// Multi-Model Dual-Path Execution Pattern in aiService.js
async function callGeminiApi(apiKey, prompt) {
  const models = ['gemini-3.6-flash', 'gemini-3.5-flash', 'gemini-1.5-flash'];
  for (const model of models) {
    try {
      const ai = new GoogleGenAI({ apiKey });
      const res = await ai.models.generateContent({ model, contents: prompt });
      if (res?.text) return res.text;
    } catch (err) {
      /* REST Fallback Retry */
    }
  }
}
```

### 2.4 API Endpoint Contract (POST /api/recommendations/generate)

Exposed via `backend/src/routes/recommendations.js`, this endpoint accepts the full project payload and returns a standardized, validated array of 4 objects.

| Field Key       | Type   | Description & Allowed Values                                 |
|-----------------|--------|--------------------------------------------------------------|
| title           | String | Short, action-oriented strategic header.                     |
| priority        | Enum   | critical   high   moderate   low                             |
| body            | String | 2–3 sentences detailing the execution plan.                  |
| impact / effort | Enum   | High   Medium   Low                                          |
| category        | String | Strategic dimension (e.g., Market Position, Unit Economics). |
| rationale       | String | 1 sentence grounding statement referencing project metrics.  |

### 2.5 Frontend UI Integration (Recommendations.jsx)

The recommendations view in `frontend/src/pages/Recommendations.jsx` renders dynamic strategy cards featuring real-time provider badges ( `GEMINI AI` , `OPENAI` , or `RULE ENGINE FALLBACK` ), animated shimmer loading skeletons, and interactive retry handlers.

### 3. Task 2: Mitigation & Improvement Suggestion Engine

#### 3.1 System Purpose & Problem Context

Identifying risks without providing remedies creates strategic friction. Task 2 establishes an analytical mitigation engine that evaluates vulnerabilities identified in Milestone 2 (e.g., high leader share, negative SOM growth, lean runway, sector compliance burdens) and generates structured tactical solutions.

#### 3.2 Multi-Dimensional Schema Architecture

Each generated mitigation record provides an end-to-end analytical breakdown across 8 mandatory dimensions:

**8-Point Mitigation Schema:**  
**riskProblem:** Explicit identification of the vulnerability. **possibleCauses:** Root-cause analysis derived from market signals. **severity:** Impact magnitude (high, medium, low). **mitigationStrategy:** Tactical defense or risk containment plan. **improvementSuggestion:** Product or business model optimization. **priority:** Execution urgency (critical, high, moderate, low). **recommendedAction:** Specific 30-day execution milestone. **expectedOutcome:** Quantitative target metric (e.g., 30% CAC reduction).

#### 3.3 Deterministic Rule Engine Synthesis

When external AI keys are unavailable, `generateFallbackMitigations()` in `mitigationEngine.js` analyzes context signals deterministically:

| Signal Condition   | Identified Risk Problem      | Synthesized Tactical Mitigation                                                               |
|--------------------|------------------------------|-----------------------------------------------------------------------------------------------|
| SOM Growth < 0%    | Contracting Sector SOM       | Deploy targeted Account-Based Marketing (ABM) on high-intent beachhead segments.              |
| Leader Share ≥ 25% | Incumbent Category Monopoly  | Position around niche workflows and 1-click migration tools from leader platform.             |
| Budget < ₹1 Crore  | Capital Runway Constraint    | Focus strictly on Core Features; structure prepaid annual contracts for non-dilutive capital. |
| Fintech / Health   | Sector Regulatory Compliance | Adopt ISO/SOC2 compliant infrastructure and embed audit trails directly into product core.    |

### 3.4 Backend Route & Execution Flow (POST /api/mitigations/generate)

The backend router `backend/src/routes/mitigations.js` handles request validation and invokes `generateMitigations()` in `backend/src/services/mitigationEngine.js`. The service enforces strict JSON output parsing, validating enum bounds and ensuring that exactly 4 comprehensive mitigation records are returned.

```
// API Endpoint Handler in mitigations.js router.post('/generate', async (req, res) => { const {
form, market, competitors } = req.body; if (!form || !form.industry) { return
res.status(400).json({ error: 'Missing required project context' }); } const result = await
generateMitigations({ form, market, competitors }); res.json(result); });
```

### 3.5 UI Panel Component (MitigationEnginePanel.jsx)

The mitigation engine is exposed to users as a dedicated card grid embedded within the Milestone 2 Risk Assessment page (`frontend/src/pages/RiskAssessment.jsx`).

#### Mitigation Engine Panel Features:

- **Severity & Priority Badges:** Color-coded tag rendering (Critical = Red, High = Orange, Moderate = Purple).
- **Root Cause & Action Callouts:** Visual distinction between underlying market causes, product improvements, and recommended 30-day actions.
- **Engine Indicator Badge:** Displays real-time execution status (Gemini AI vs Rule Engine Fallback).
- **Seamless Context Sync:** Automatically updates whenever the active industry or project budget is modified in the intake form.

### 3.6 Rate Limiting & Resilience

If external AI APIs return rate limits (HTTP 429) or connection timeouts, the mitigation engine catches the exception silently, logs the diagnosis internally, and returns rule-derived mitigations to preserve 100% frontend uptime.

## 4. Task 3: LangGraph Multi-Agent Agent Workflow

### 4.1 Framework Architecture (@langchain/langgraph)

Task 3 introduces autonomous agent orchestration using **LangGraph** (@langchain/langgraph). Unlike standard sequential API chains, LangGraph models execution as a cyclic StateGraph with explicit state definitions, node handlers, and conditional transition edges.



### 4.2 Detailed 8-Node State Machine Breakdown

The workflow graph defined in backend/src/services/langgraphWorkflow.js initializes a centralized state channel containing 8 specialized agent nodes:

| Node Name               | Primary Responsibility & State Mutation                                                         |
|-------------------------|-------------------------------------------------------------------------------------------------|
| dataAnalysis            | Parses raw TAM/SAM/SOM growth rates and competitor share distributions into analytical signals. |
| riskIdentification      | Scans for critical flags (e.g., SOM < 0%, leader share > 25%) and sets hasCriticalRisks flag.   |
| strategicRecommendation | Synthesizes 4 data-grounded strategic positioning pillars.                                      |
| mitigationAnalysis      | Formulates tactical defenses and risk containment strategies.                                   |
| improvementSuggestions  | Generates product optimizations and business model refinements.                                 |
| validationReview        | Evaluates overall dossier quality score (0–100) based on completeness and grounding.            |
| refinement              | Triggered conditionally if quality score < 70; enhances action items and rationale cards.       |
| finalSynthesis          | Assembles complete strategic dossier and compiles execution step log.                           |

### 4.3 Conditional Edge Routing Mechanics

The power of LangGraph lies in dynamic control flow. The workflow incorporates two conditional routing functions:

#### Conditional Routing Rules:

- **Priority Risk Routing (`routeAfterRisk`):** If `hasCriticalRisks === true`, the graph immediately routes execution directly to `mitigationAnalysis` to prioritize risk containment before general strategy synthesis.
- **Quality Audit Refinement (`routeAfterValidation`):** Following `validationReview`, if the computed `qualityScore < 70`, the state machine branches to `refinement` to enrich rationale clarity before advancing to `finalSynthesis`.

### 4.4 Backend Route (POST `/api/workflow/run`)

Exposed via `backend/src/routes/workflow.js`, this route executes the compiled `StateGraph` asynchronously, tracking step execution timestamps and returning the final dossier alongside an execution log array.

### 4.5 Visual Pipeline Viewer (`LangGraphWorkflowViewer.jsx`)

Integrated into the Recommendations page (`frontend/src/pages/Recommendations.jsx`), the `LangGraphWorkflowViewer` provides a live UI visualization of the 8-node state machine:

#### Visual Component Capabilities:

- **Node State Badges:** Real-time visual status icons for active, completed, and refined graph nodes.
- **Quality Score Gauge:** Displays the dossier audit score computed during `validationReview`.
- **Execution Step Trace Log:** Expandable audit log listing node entry times and state mutation outputs.

## 5. Verification, Security & Production Deployment

### 5.1 Empirical Verification Matrix

Milestone 3 was subjected to automated build and syntax verification commands to confirm zero regression across Milestones 1 and 2:

| Verification Area         | Command Executed                          | Result & Evidence                                             |
|---------------------------|-------------------------------------------|---------------------------------------------------------------|
| Frontend Production Build | <code>npm run build</code> (in frontend/) | <b>PASS</b> — 67 modules transformed in 465ms with 0 errors.  |
| Backend Syntax & Imports  | <code>node --check src/server.js</code>   | <b>PASS</b> — Syntax check passed cleanly with 0 errors.      |
| Task 1 AI Recommendations | Automated End-to-End Suite                | <b>PASS</b> — Validated 4 items with provider status badges.  |
| Task 2 Mitigation Engine  | Automated End-to-End Suite                | <b>PASS</b> — Validated 4-point mitigations & severity tags.  |
| Task 3 LangGraph Workflow | Automated End-to-End Suite                | <b>PASS</b> — Validated 8-node state machine & audit routing. |

### 5.2 Security Posture & Secrets Management

- **Zero Client Exposure:** GEMINI\_API\_KEY and OPENAI\_API\_KEY reside strictly in backend/ .env and are never bundled into client JS assets.
- **Sanitized Errors:** Server stack traces are logged internally while sending safe JSON responses to the frontend.

### 5.3 Conclusion & Sign-Off

Milestone 3 completes the transformation of LaunchLens into an intelligent, data-grounded startup strategic advisor. The integration of multi-provider AI, structured mitigation synthesis, and LangGraph agent workflows provides enterprise-grade analytical capabilities ready for production deployment.

**Prepared By:** Engineering Team  
**Approved By:** System Architect

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